

Probability = How likely something is to happen.

👉 Statistics = How we collect, understand, and learn from data.

probability value means 0 to 1

A random variable is a variable whose value depends on the outcome of a random experiment.



You roll a die → possible outcomes: 1, 2, 3, 4, 5, 6.

Let

X = number that appears on the top.

Here, X is the random variable.

We don't know the value before rolling → that's why it's random.

graph text

probability distribution

A probability distribution basically tells us how probabilities are spread across all possible outcomes of a random event .

What is Discrete Probability Distribution? A discrete probability distribution shows all the possible values a random variable can take and how likely each value is.

- “Discrete” → the values are separate, countable, not continuous.
- Probabilities always add up to 1.

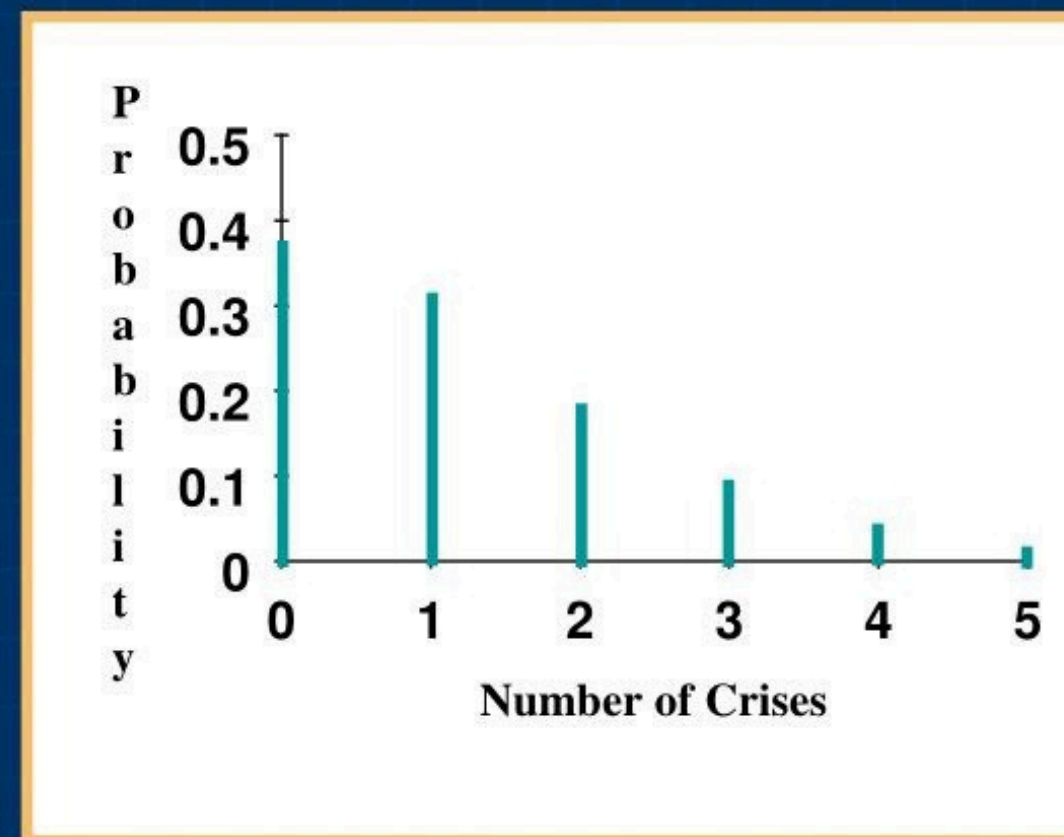
Think: like a bar chart of chances.

Tossing a coin

- $X = 1$ if Head, 0 if Tail
- $P(X=1) = 0.5, P(X=0) = 0.5$

Example: Discrete Distributions & Graphs

Distribution of Daily Crises	
Number of Crises	Probability
0	0.37
1	0.31
2	0.18
3	0.09
4	0.04
5	0.01





Coca Cola



Coffee



Egg (E.G.G)



Milk



Pizza



Sausages



Sugar



Orange juice

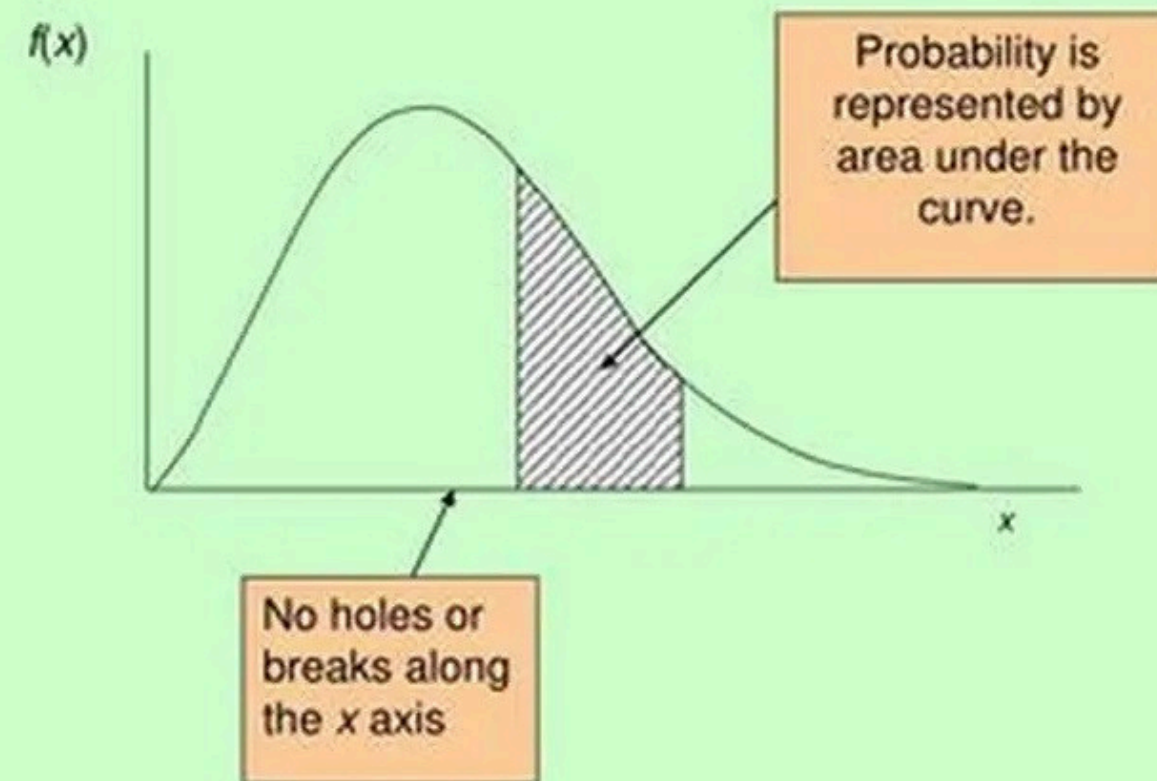


Tea

A continuous probability distribution describes the probabilities of a continuous random variable, which can take any value within a range.

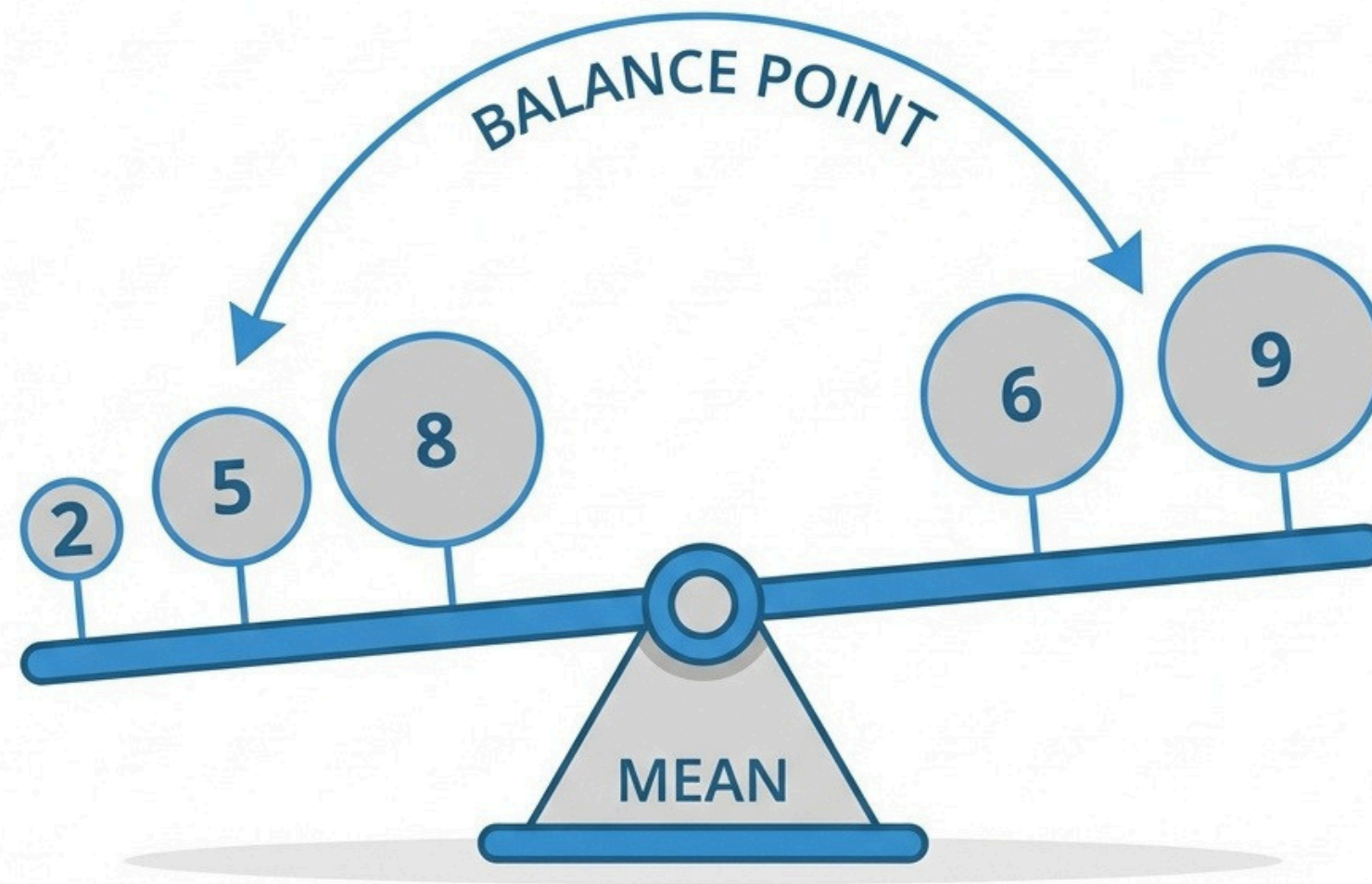
- Unlike discrete variables (countable outcomes like dice), continuous variables are infinite and uncountable.
- Probabilities are measured using areas under a curve instead of exact values.

Figure 6.2 A Continuous Probability Distribution



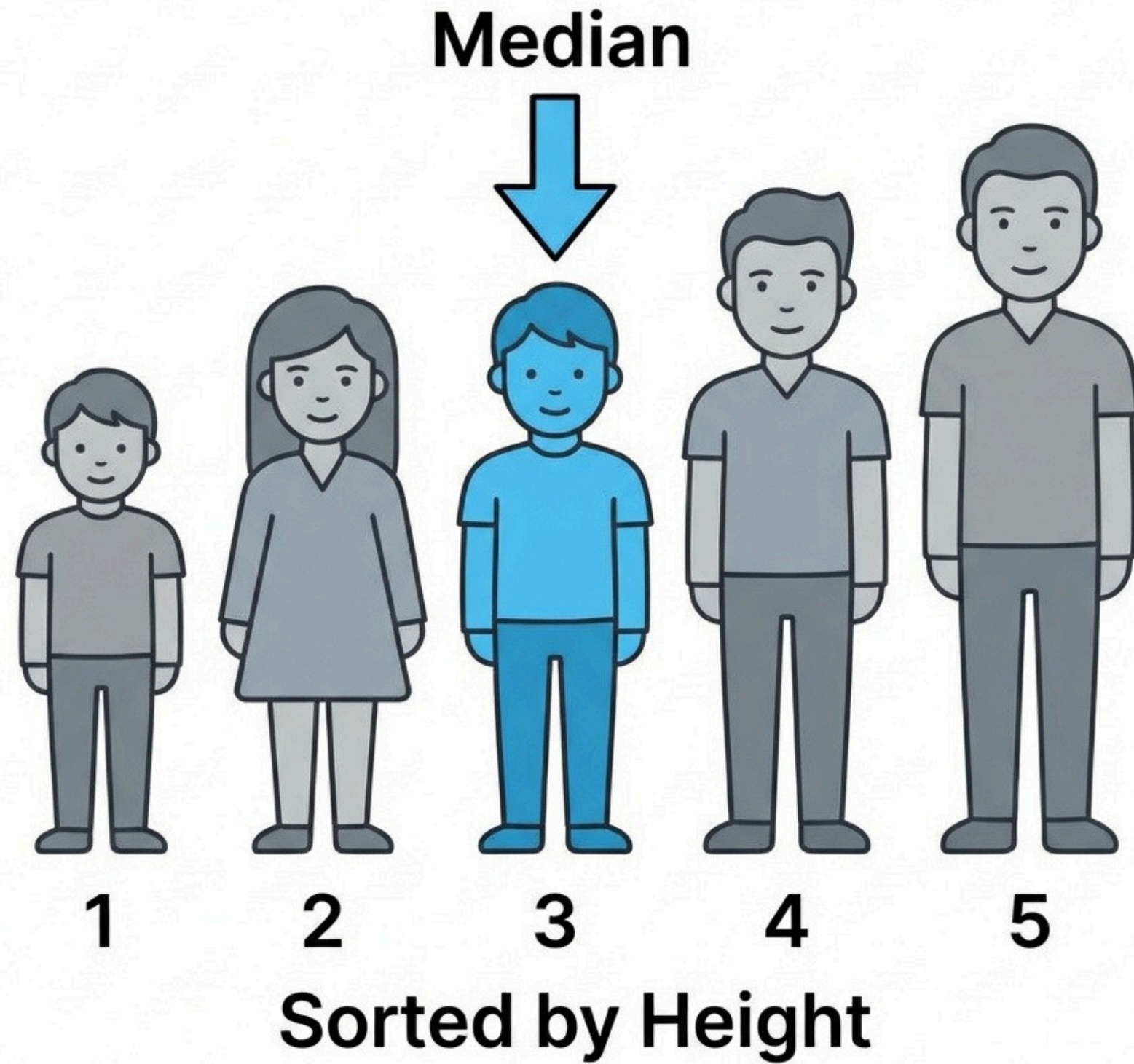
'Mean'

(Arithmetic Average)

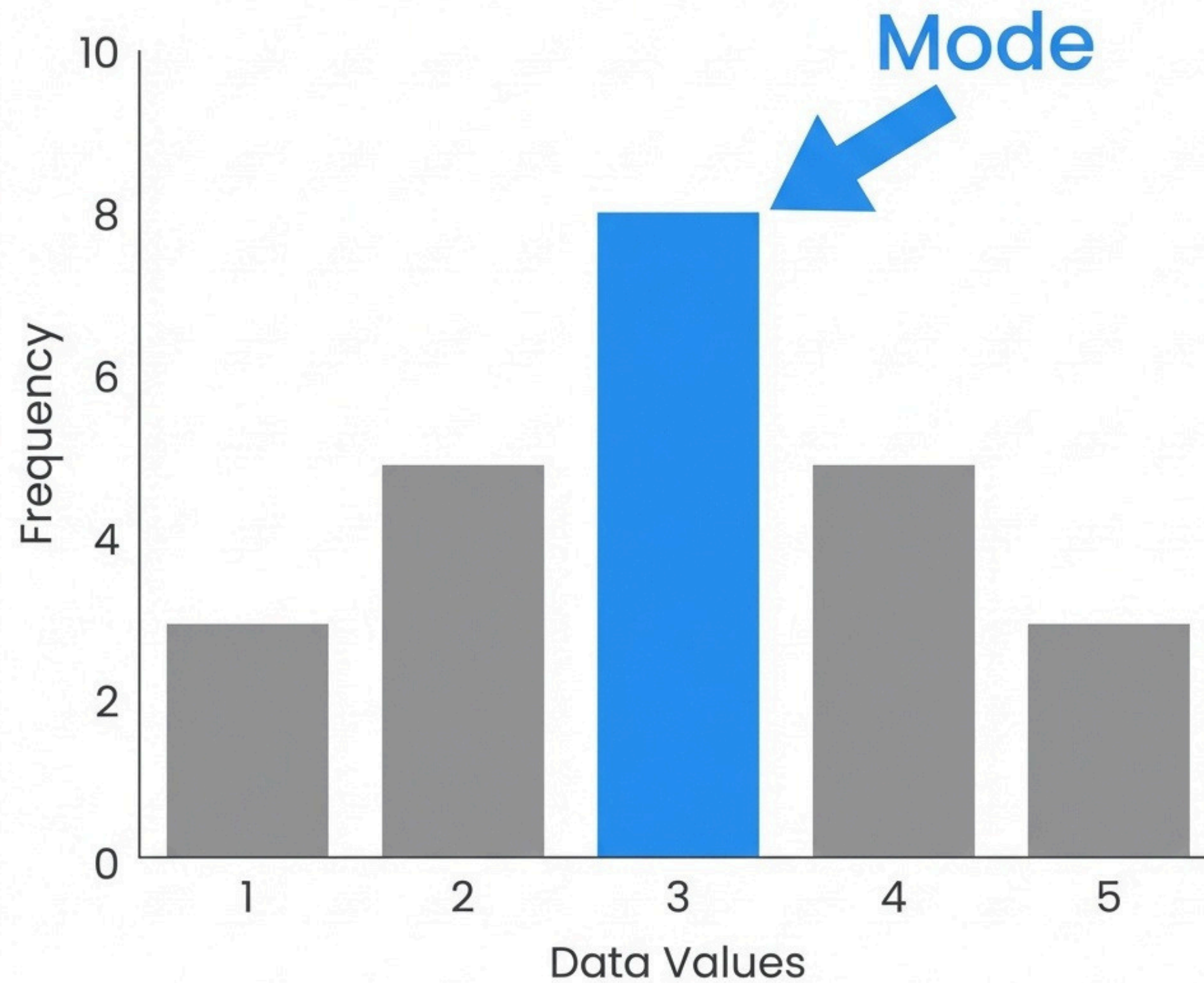


$$\text{Mean} = (2+5+8+6+9) / 5 = 30 / 5 = 6$$

What is the Median?



Understanding Mode: The Most Frequent Value



Variance

Data: [2, 4, 6, 8]

1. Mean: $\bar{x} = (2 + 4 + 6 + 8)/4 = 5$

2. Deviations: $[2-5, 4-5, 6-5, 8-5] = [-3, -1, 1, 3]$

3. Squared deviations: $[9, 1, 1, 9]$

4. Variance (population):

$$\sigma^2 = \frac{9 + 1 + 1 + 9}{4} = \frac{20}{4} = 5$$

- So variance = 5, meaning the numbers are spread out from the mean.

What is Standard Deviation (σ)?

- Standard deviation measures how spread out data is from the mean.
- Smaller $\sigma \rightarrow$ data is closer to the mean
- Larger $\sigma \rightarrow$ data is more spread out

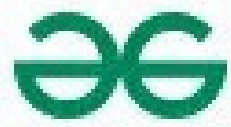
Think of it as:

“How tightly the data points hug the mean.”

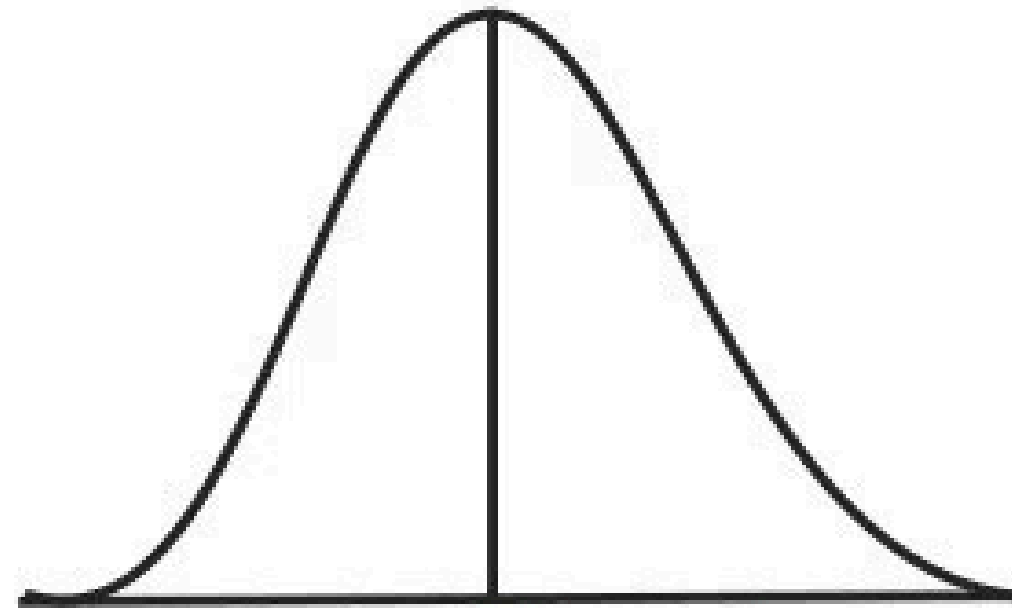
What is Skewness?

Skewness measures the asymmetry of a data distribution.

- It tells us whether the data is tilted to the left, right, or balanced.
- Basically:
 - Is the data symmetrical?
 - If not, which side has the “long tail”?

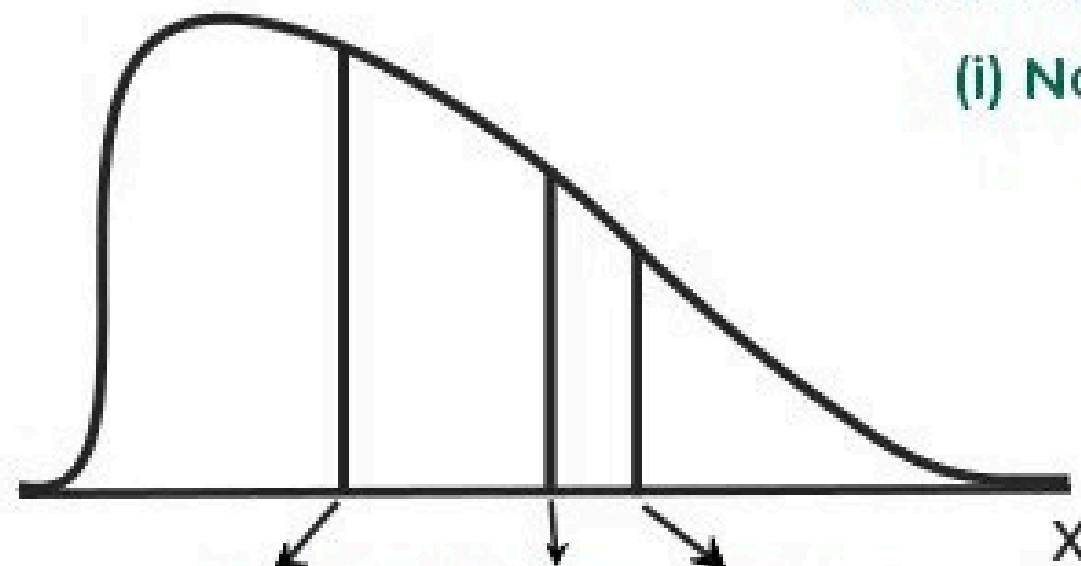


Skewness



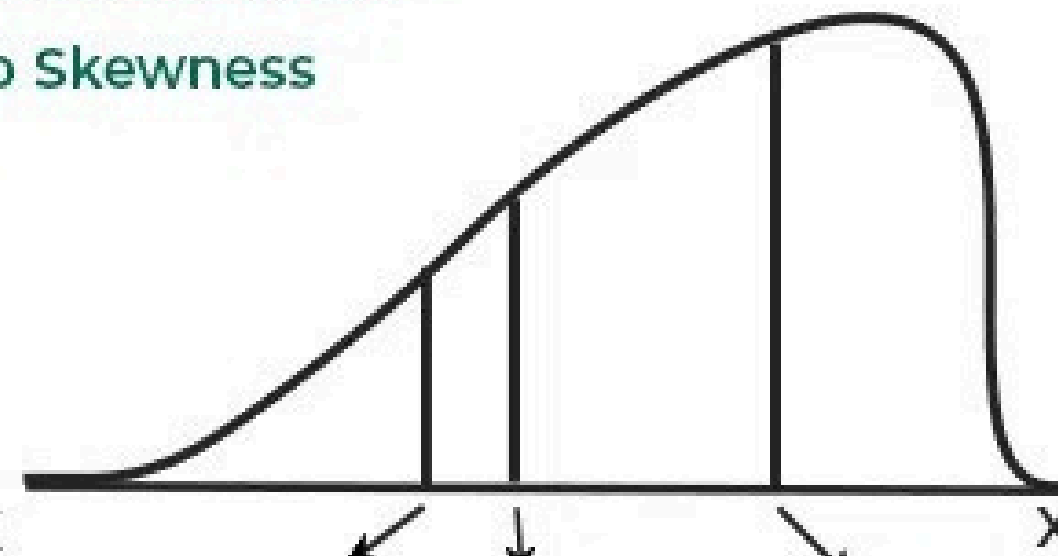
Mean= Median= Mode

(i) No Skewness



Mode Median Mean

(ii) Positive Skewness



Mean Median Mode

(iii) Negative Skewness

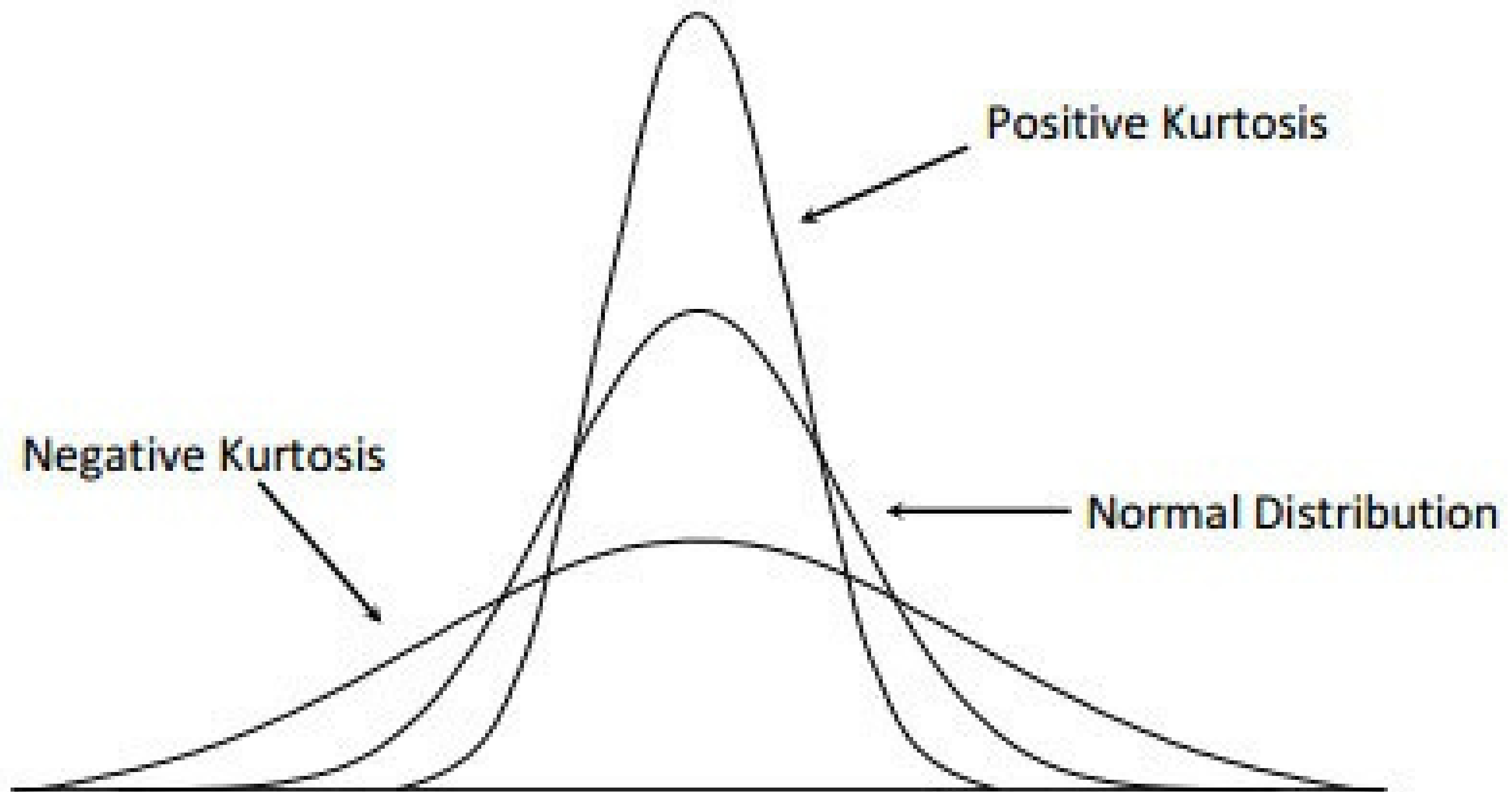
What is Kurtosis?

Kurtosis measures the “peakedness” or “flatness” of a probability distribution.

- It tells us how heavy or light the tails of the distribution are compared to a normal distribution.

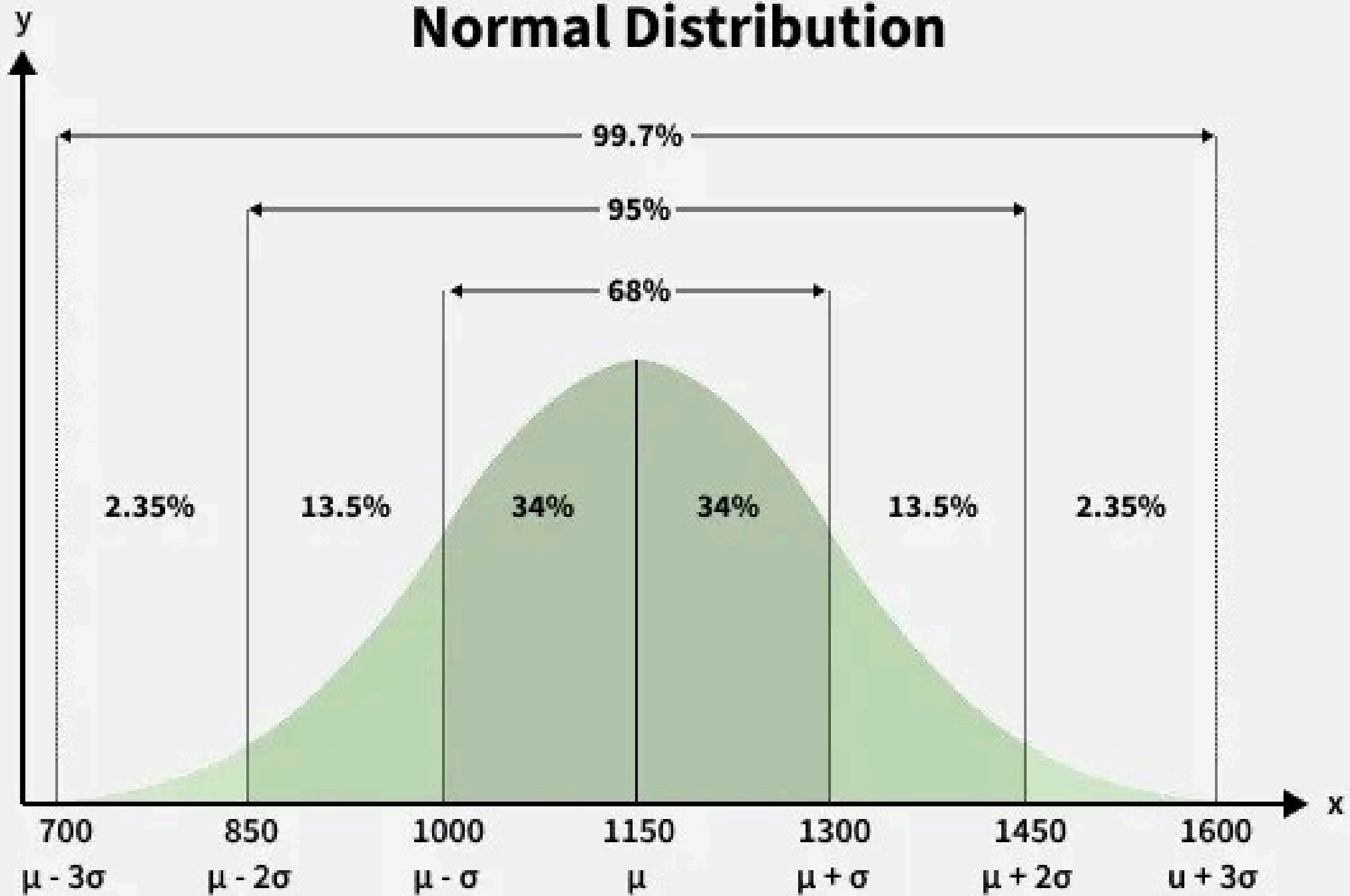
In simple words:

It shows whether the data has more extreme values (outliers) or is flatter.



- Normal distribution $\rightarrow$ kurtosis ≈ 3 (mesokurtic) Dataset with many
- extreme high/low values $\rightarrow$ kurtosis > 3 (leptokurtic) Dataset with evenly spread values $\rightarrow$
- kurtosis < 3 (platykurtic)

Normal Distribution



- Many natural phenomena follow it (heights, weights, errors).
- Symmetric $\rightarrow$ mean = median = mode $\rightarrow$ easy analysis.
- Basis of statistical tests (t-test, ANOVA, regression).
- Predictable (68–95–99.7 rule) $\rightarrow$ understand spread and outliers.
- Central Limit Theorem $\rightarrow$ sums/averages of variables $\rightarrow$ normal.
- Widely used in ML $\rightarrow$ Gaussian Naive Bayes, probabilistic models, density estimation.

1 25th Percentile (Q1)

- Also called **first quartile (Q1)**.
- **25% of the data is less than this value, 75% is more.**
- Think of it as the “**lower boundary of the lower quarter**”.

Example:

Data (sorted): 10, 12, 15, 18, 20, 22, 25, 30

- 25th percentile $\approx 14 \rightarrow$ **25% of numbers are ≤ 14**

2 75th Percentile (Q3)

- Also called third quartile (Q3).
- 75% of the data is less than this value, 25% is more.
- Think of it as the “upper boundary of the upper quarter”.

Example (same data):

- 75th percentile $\approx 24 \rightarrow 75\%$ of numbers are ≤ 24

2 Using IQR (Interquartile Range) – Very Popular

- Step 1: Find Q1 (25th percentile) and Q3 (75th percentile)
- Step 2: Calculate IQR:

$$IQR = Q3 - Q1$$

- Step 3: Define outlier limits:

$$\text{Lower limit} = Q1 - 1.5 \times IQR$$

$$\text{Upper limit} = Q3 + 1.5 \times IQR$$

- Data outside these limits = outliers